

Perceived Benefits and Risks of Information Disclosure in Augmented Reality

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Figure 1: The three theoretical scenarios explored in the study: a pub, a campus, and a conference

ABSTRACT

By merging digital content with the physical environment, AR offers unique social interaction opportunities alongside distinct risks. This study explores perceived benefits and risks of sharing personal info in AR social contexts. Analysis of participant interviews, conducted after presenting three AR scenarios, revealed benefits like targeted communication, networking opportunities, and fostering shared values, varying by scenario. Participants also identified risks like prejudgement, harassment, and diminished emotional connection, also shown to be context-dependent. These findings demonstrate the dynamic interplay between advantages and vulnerabilities of self-disclosure in AR, highlighting the need for systems addressing these concerns.

Index Terms: Augmented reality, Self disclosure, Self presentation, Technology mediated social interaction

1 INTRODUCTION

Identity and self-presentation are inherently fluid and context-dependent, shaped by individual goals and social environments [1, 4, 3]. In today's digital age, social interaction often involves mediated communication, where the decision to disclose personal information is influenced by the unique characteristics of digital environments, which often highly differs from face-to-face interactions [5]. Social interactions now span the entirety of the virtuality-reality continuum. This includes experiences ranging from physical reality to fully immersive virtual environments [6]. These shifting contexts may profoundly influence self-disclosure, as the degree of virtuality determines how individuals navigate privacy, audience perception, and the intimacy of shared information. As digital interactions increasingly blend the physical and virtual worlds, the boundaries between online and offline self-presentation grow increasingly complex, raising new questions about privacy and identity in these hybrid spaces.

AR integrates digital content into real-world environments through devices such as smart glasses, creating immersive experiences that challenge users to rethink privacy and identity. For

instance, AR glasses in professional networking settings may display personal profiles in real-time, facilitating connections on the one hand but raising privacy concerns on the other. Studies on AR and social interaction suggest that AR's immersive and probing nature can decrease comfort levels, particularly compared to screen-based interactions such as smartphone-based communication. Consequently, AR's merging of virtual and real-world content challenges users to address privacy and self-presentation in unique ways. These challenges are notably acute in shared physical environments, where any disclosures become linked to the user's actual identity. Through scenario-based interviews, this study investigates the perceived benefits and risks of self-disclosure in AR across diverse social contexts, providing insights into how AR systems can better support user needs.

2 METHODS

The study consisted of 15 participants whose ages ranged from 25 to 63 years, with a mean age of 36.2 years and a standard deviation of 9.8 years. Of these participants, six identified as female, whereas nine identified as male.

Participants were introduced to augmented reality through a demo created with Meta Spark Studio. The demo showcased how personal information could be displayed in AR, providing participants with a clear understanding of how the technology functions (see Figure 2).



Figure 2: Fictional profiles and identities of participants.

They were then briefed on three theoretical scenarios where AR would be used as follows:

- The Pub: "You and your friends went to a bar in the evening; before entering the pub, you were asked if you would like to join an AR network in the pub. If you choose to join, you will be given AR glasses to scan other participants in the pub. You will be asked to fill out a form using your smartphone to disclose information about yourself that the other participants in the network could see".

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- The Conference: “You have been invited to a private conference about your job or field of study. Upon entering the conference, you are asked to join an AR network. If you choose to join, you will be given AR glasses that can scan other participants in the conference and see information on these participants. You will be asked to fill out a form using your smartphone in which you will disclose information about yourself that the other participants in the network could see.
- The Campus: “You are a student at a university. You are asked to join an AR network. If you choose to join, you will be given AR glasses that can scan other participants on the campus, and you will be asked to fill out a form using your smartphone in which you will disclose information about yourself”.

Finally, a semi-structured interview was conducted with each participant about their willingness to disclose personal information and their perceptions of the benefits and risks of doing so in each of the three scenarios.

The data analysis procedure consisted of thematic analysis as described in [2]. The interviews were transcribed and then imported into the Atlas.ti software. Using a hybrid coding approach, we combined top-down and bottom-up methods to explore themes comprehensively, leveraging the strengths of each approach.

3 PRELIMINARY RESULTS

First, it was found that most participants were willing to disclose information to some extent, with the highest willingness observed in the conference scenario (66.2%), followed by the campus (53.3%) and pub (50.8%) scenarios.

Table 1: Willingness to disclose personal information

	Pub	Campus	Conference
Yes	50.8%	53.3%	66.2%
No	49.2%	46.7%	33.8%

Next, participants described what they perceive to be the benefits and risks of disclosing personal information in AR. Benefits refer to what one hopes to gain from such an interaction, while risks refer to the possibility of negative consequences resulting from the self-disclosure of personal information.

3.1 Benefits of disclosing information

Table 2: Number of comments identifying a specific benefit / proportion out of 100% per scenario

Benefit	Pub	Campus	Conference
Sharing Contact Info	2 / 4.8%	0 / 0.0%	8 / 19.0%
Informed Interaction	12 / 28.6%	0 / 0.0%	10 / 23.9%
Targeted Communication	20 / 47.7%	4 / 28.6%	14 / 33.3%
Shared Values and Interests	8 / 19.0%	10 / 71.4%	10 / 23.9%

The benefits identified include sharing contact information for future networking, informed interactions through background sharing, targeted communication to find individuals with common goals, and expressing shared interests to foster deeper connections. What unites these benefits is their role in improving the quality of social exchanges by making them more relevant, efficient, and meaningful. The findings indicate that in settings like a conference, the range of benefits could lead to a broader disclosure, as the benefits here are numerous and not concentrated on a singular significant benefit. In contrast, pursuing a specific benefit, such as ‘shared values and interests’, in a pub setting might result in a more focused and narrow disclosure.

Table 3: Number of comments identifying a specific risk / proportion out of 100% per scenario

Risk	Pub	Campus	Conference
Fear of Prejudgment	5 / 17.2%	5 / 33.3%	4 / 17.4%
Inappropriate Disclosure	1 / 3.4%	4 / 26.7%	12 / 52.2%
Information Getting to Unintended Places	2 / 6.9%	3 / 20.0%	3 / 13.0%
Potential for Harassment	11 / 37.9%	3 / 20.0%	1 / 4.3%
Concerns Regarding Spam	8 / 27.6%	0 / 0.0%	1 / 4.3%
Reduced Emotional Connection	2 / 6.9%	0 / 0.0%	2 / 8.7%

3.2 Risks of disclosing information

With respect to risks, participants expressed concerns about harassment based on their shared information, particularly in the pub scenario, which saw the highest incidence of such worries. They also feared the discomfort of revealing inappropriate information, a risk most prevalent in the conference scenario due to its professional context. In addition, the potential for personal information to fall into the hands of unintentional recipients, including spam companies, was a concern in all scenarios, implying privacy concerns. Another concern raised was the risk of real-life harassment, accentuated by AR’s unique blend of digital and physical worlds. This concern was pronounced in the pub setting, reflecting its public and unrestrained nature. Furthermore, our results highlight a specific concern related to the nature of interactions in AR: the risk of reduced emotional connection. Participants feared that an overreliance on digitally mediated communication in AR might diminish the depth and authenticity of their interactions, indicating a potential loss in the quality of emotional connections.

4 CONCLUSION AND FUTURE DIRECTIONS

This study highlights the unique circumstances of self-disclosure in AR, where benefits such as targeted communication and fostering connections coexist with significant risks such as privacy breaches, harassment, and reduced emotional depth. The general willingness of the participants to disclose varied between the scenarios, as well as the benefits and risks perceived by the disclosure of information in the different scenarios. This emphasizes the importance of context in shaping perceived benefits and vulnerabilities. These findings underscore the need for AR systems to balance the opportunities for enhanced social interactions with robust mechanisms for privacy and control, ensuring that users can navigate AR environments with both confidence and agency.

REFERENCES

- [1] R. F. Baumeister and D. G. Hutton. Self-Presentation Theory: Self-Construction and Audience Pleasing. In B. Mullen and G. R. Goethals, eds., *Theories of Group Behavior*, pp. 71–87. Springer, New York, NY, 1987. doi: 10.1007/978-1-4612-4634-3_4 1
- [2] V. Braun and V. Clarke. Using thematic analysis in psychology. *Qualitative research in psychology*, 3(2):77–101, 2006. Publisher: Taylor & Francis. 2
- [3] S. Gangestad and M. Snyder. Self-Monitoring: Appraisal and Reappraisal. *Psychological bulletin*, 126:530–55, July 2000. doi: 10.1037/0033-2909.126.4.530 1
- [4] M. A. Hogg. Social Identity Theory. In S. McKeown, R. Haji, and N. Ferguson, eds., *Understanding Peace and Conflict Through Social Identity Theory: Contemporary Global Perspectives*, pp. 3–17. Springer International Publishing, Cham, 2016. doi: 10.1007/978-3-319-29869-6_1 1
- [5] J. Kim and K. Dindia. Online self-disclosure: A review of research. *Computer-mediated communication in personal relationships*, pp. 156–180, 2011. 1
- [6] P. Milgram and F. Kishino. A taxonomy of mixed reality visual displays. *IEICE TRANSACTIONS on Information and Systems*, 77(12):1321–1329, 1994. Publisher: The Institute of Electronics, Information and Communication Engineers. 1